

ABOUT ME

Hi, I'm Okik an aspiring AI and Data Science engineer passionate about building intelligent systems that solve real-world problems.

I'm currently studying Agricultural Engineering at the Federal University of Agriculture, Abeokuta (FUNAAB), while actively developing my skills in machine learning, artificial intelligence, and software development. Over time, I've worked on projects ranging from customer churn prediction systems to AI-powered agriculture solutions and conversational AI applications.

What excites me most about AI is the ability to turn data into useful, interactive, and impactful systems. I enjoy building projects that combine practical problem-solving with modern AI technologies, especially in areas like machine learning, intelligent automation, smart agriculture, and large language model applications.

Beyond building models, I'm also interested in the full AI development process from data preprocessing and feature engineering to deployment and user experience. I enjoy learning by building real projects and continuously exploring new technologies that improve how intelligent systems interact with people.

Currently, I'm expanding my knowledge in Retrieval-Augmented Generation (RAG), vector databases, conversational AI systems, and modern AI engineering workflows while growing a portfolio focused on practical and scalable AI applications.

EXPERIENCE

Over the past few years, I have been actively building hands on experience in machine learning, artificial intelligence, and data science through personal projects, online learning, and practical experimentation.

My learning journey has been heavily self-driven, with a strong focus on applying concepts through real-world projects rather than only studying theory. I have spent time working on machine learning workflows involving data preprocessing, feature engineering, model training, evaluation, and deployment using Python-based tools and frameworks.

In addition to developing personal AI projects, I have also worked as a freelance data science tutor, helping beginners learn Python, data analysis, and machine learning concepts through practical examples and project-based teaching.

I enjoy learning by building, solving technical problems, and continuously improving my understanding of modern AI systems and development workflows.

EDUCATIONAL BACKGROUND

I am currently studying Agricultural Engineering at the Federal University of Agriculture, Abeokuta (FUNAAB), Nigeria. Alongside my university education, much of my knowledge in artificial intelligence, machine learning, and software development has been developed through self-learning. I have completed multiple online courses and followed practical learning resources from platforms such as Udemy, YouTube, and developer documentation.

My learning approach focuses strongly on hands-on practice, project development, and continuous experimentation with real-world AI applications.

PROJECT

Throughout my learning journey, I have worked on several machine learning and AI-focused projects designed to solve practical problems and strengthen my understanding of real-world AI systems. My projects mainly focus on predictive analytics, intelligent agriculture systems, conversational AI, and machine learning deployment.

Project I have worked on include: Customer Churn Prediction System, Smart Irrigation Prediction System, Crop Pest Detection System, Smart Agriculture AI System, AI Portfolio RAG Chatbot

PROJECT: CUSTOMER CHURN PREDICTION SYSTEM

The Customer Churn Prediction System is a machine learning project developed to help businesses identify customers who are likely to stop using their services. The goal of the system is to support customer retention by providing early insights into churn behavior and helping businesses make more informed decisions.

The project involved building a complete machine learning workflow, including data preprocessing, feature engineering, model training, evaluation, and deployment preparation. During development, the initial model achieved an accuracy of 92%. After applying feature selection techniques to reduce model complexity and improve efficiency, the model maintained a strong accuracy of 90% while becoming lighter and more optimized.

To make the system more practical for real business use cases, customer predictions were grouped into three categories based on churn probability:

High Risk Churn Customers

Medium Risk Churn Customers

Low Risk Churn Customers

This classification approach allows businesses to prioritize customer retention strategies more effectively instead of relying only on a simple churn or non-churn prediction.

The project also focused on creating an interactive and user-friendly prediction workflow that could easily integrate into a business environment.

Technologies and tools used in this project include Python, Pandas, Scikit-learn

PROJECT: SMART IRRIGATION PREDICTION SYSTEM

The Crop Pest Detection System is a deep learning and computer vision project developed to identify crop pests using image classification techniques, with the goal of supporting early detection of agricultural threats.

Unlike relying solely on pre-trained solutions, this project was trained with a more controlled approach to improve understanding of the model behavior and achieve better performance tuning. This allowed for more direct control over the training process, dataset handling, and model optimization.

A key focus during development was reducing overfitting and improving generalization. To achieve this, early stopping techniques were implemented to monitor validation performance and prevent the model from over-training once performance stopped improving. This helped improve stability and overall reliability of the model.

In addition, the trained model was converted to ONNX format to improve inference speed and deployment efficiency. This optimization step made the model lighter and more suitable for real-time or production-like environments.

The project involved image preprocessing, dataset preparation, model training, evaluation, and performance tuning across different pest categories.

Through this project, I gained practical experience in deep learning workflows, computer vision, model optimization, and deployment-aware machine learning development.

Technologies and tools used include Python, TensorFlow, Keras, Computer Vision techniques, and ONNX for model optimization.

PROJECT: CROP PEST DETECTION SYSTEM

The Crop Pest Detection System is a deep learning and computer vision project developed to identify crop pests and support early agricultural disease monitoring through image classification techniques.

Unlike using a fully pre-trained solution, this project involved training the model more directly and intentionally to achieve better control over the learning process, model behavior, and overall prediction performance. This approach provided deeper hands-on experience with image classification workflows, model optimization, and training strategies within deep learning systems.

A strong focus during development was improving model generalization and reducing overfitting. To achieve this, early stopping techniques were implemented during training to monitor validation performance and automatically stop training when improvements began to decline. This helped improve model stability while preventing unnecessary over-training.

The project also involved image preprocessing, dataset preparation, model evaluation, and performance tuning to improve classification accuracy across different crop pest categories.

Through this project, I gained practical experience working with deep learning pipelines, image-based datasets, model training strategies, and computer vision concepts applied to real-world agricultural problems.

Technologies and tools used in this project include Python, TensorFlow, Keras, Computer Vision techniques, and Streamlit.

PROJECT: SMART AGRICULTURE AI SYSTEM

The Smart Agriculture AI System is an integrated machine learning project that combines multiple agricultural AI solutions into a unified intelligent system. It was designed to support decision-making in agriculture by merging predictive irrigation insights with crop pest detection capabilities.

This project brings together the Smart Irrigation Prediction System and the Crop Pest Detection System into one cohesive AI-driven platform. The idea behind it was to demonstrate how multiple machine learning models can work together to solve related real-world problems within a single ecosystem.

On one side, the system provides irrigation recommendations based on environmental and soil conditions, helping improve water usage efficiency in farming. On the other side, it incorporates an image-based crop pest detection component that identifies potential pest-related issues affecting crops.

The main focus of this project was system integration, where different machine learning models were combined into a single workflow that can handle multiple agricultural tasks. It also helped in understanding how AI systems can be structured beyond individual models to create more complete and practical solutions.

This project strengthened my experience in building modular AI systems, integrating multiple machine learning pipelines, and designing end-to-end intelligent applications for real-world use cases.

Technologies and tools used include Python, Scikit-learn, TensorFlow, Computer Vision techniques, and Streamlit.

PROJECT: AI PORTFOLIO RAG CHATBOT

The AI Portfolio RAG Chatbot is a retrieval-augmented generation system built to serve as an interactive assistant for my personal portfolio. Instead of being a static website, the portfolio is enhanced with a conversational AI that allows users to ask questions about my skills, projects, experience, and contact information.

The system is designed to retrieve relevant information from a structured knowledge base containing details about me and my work, and then use a language model to generate clear, context-aware responses. This makes the portfolio more dynamic, engaging, and easier to explore for recruiters or collaborators.

The core of the project is a RAG (Retrieval-Augmented Generation) pipeline, where user queries are converted into embeddings and matched against a vector database to retrieve the most relevant context. This context is then passed to a language model to generate accurate and grounded responses.

The project was built with a focus on scalability and real-world AI system design, including proper document structuring, chunking strategies, and semantic search optimization. It also demonstrates how large language models can be combined with vector databases to create intelligent, context-aware applications.

Through this project, I gained practical experience in RAG architecture, embeddings, vector databases, and building conversational AI systems that go beyond simple chatbot interactions.

Technologies and tools used include Python, LangChain, ChromaDB, vector embeddings, and large language model APIs.

CONTACT

I'm always open to connecting with people in tech, collaborating on interesting projects, or discussing opportunities in AI, machine learning, and data science.

Whether it's a question, a project idea, or a professional opportunity, feel free to reach out. I usually respond as quickly as possible and enjoy meaningful conversations around technology and problem-solving.

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